europass Curriculum vitae

## PERSONAL INFORMATION

# Mihai-Cristian Farcaș




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Date of birth 5 April 2004 | Nationality Romanian

## WORK EXPERIENCE

### 2026 – Present — Software Engineer (Freelance), BluePlan

Built and shipped Atlas (blueplan.vercel.app) end to end: retrieval-augmented chat over technical documents with hybrid dense, sparse and image retrieval fused with RRF, reranking and verbatim citations pinned to the source PDF. Multi-tenant, multilingual and in daily production use.

Technologies Python, FastAPI, Qdrant, React

### 2026 — Full-Stack Developer, Baseline Studios

Delivered full-stack features across multiple client applications (stratosphere.vip, pursuitofx.org, kabu.xyz, esoftplus-landing.vercel.app) using TypeScript and React.

Technologies TypeScript, React

### 2025 — Software Developer, GeoDraw

Developed an AI-powered canvas for sketching and wireframing apps with React, TLDraw and LLM APIs, turning natural language into accurate real-time design output on the canvas.

Technologies React, TLDraw, LLM APIs

### April – June 2025 — Software Development Intern, Tvarita

Rebuilt and redesigned a server monitoring site for an optician clinic chain using React and Tailwind CSS, improving the user experience and significantly reducing load times.

Technologies React, Tailwind CSS

## EDUCATION AND TRAINING

### 2026 – Present — MSc in Cloud, Network Infrastructure and High-Performance Computation

Babeș-Bolyai University, Faculty of Mathematics and Computer Science, Cluj-Napoca, Romania

### Oct 2023 – June 2026 — BSc in Computer Science (English)

Babeș-Bolyai University, Faculty of Mathematics and Computer Science, Cluj-Napoca, Romania

Thesis: Hybrid post-quantum key exchange for WireGuard, the first FIPS 203-compliant hybrid WireGuard implementation on a production Rust codebase.

## PERSONAL PROJECTS

### 2026 Helio

Open-source remote desktop and game streaming over a post-quantum WireGuard tunnel: one native app acts as client and host on macOS, Windows and Linux, with ML-KEM-768 encryption, NAT traversal through the BitTorrent DHT, Reed-Solomon loss recovery and HDR video at up to 120 fps.
helio.mihaicristian.dev

Technologies Rust, GPUI, ML-KEM-768, WireGuard, AV1/HEVC

### 2026 Atlas

AI assistant for technical documents (PDF, Word, presentations and e-books) that answers in the user's language with every claim tied to a citation opening the exact highlighted passage; hybrid dense, sparse and image retrieval with RRF fusion and reranking, multi-tenant and in daily production.
atlas-ai.ro

Technologies Python, FastAPI, Qdrant, React, LLM APIs

### 2026 asides

Screen-annotation tool for reporting bugs out loud: freeze any app, pin what is wrong and say why; local Whisper transcription types the aside into a coding agent's chat as a message with the annotated screenshot and measured geometry.
asides.mihaicristian.dev

Technologies Rust, GPUI, Whisper, MCP

### 2026 pq-boringtun

Hybrid post-quantum key exchange for WireGuard: ML-KEM-768 alongside X25519 in the Noise IK handshake, built in Rust on Cloudflare's BoringTun and benchmarked on Apple M4 and Raspberry Pi 5.
github.com/mihaicristianfarcas/pq-boringtun

Technologies Rust, ML-KEM-768, X25519, Noise Protocol, WireGuard

### 2026 Supernova

Live-wallpaper engine for macOS in Swift 6: renders looping video and sandboxed web bundles behind desktop icons, with audio, cursor and motion reactivity, per-display assignment and smart playlists.
supernova.mihaicristian.dev

Technologies Swift 6, Metal, Core Audio

### 2025 eBPF Network Analysis System

Network traffic classification for QUIC protocols, inspecting encrypted flows at the kernel boundary using eBPF with C, Rust and Python.
github.com/mihaicristianfarcas/ebpf-network-analysis-system

Technologies C, Rust, Python, eBPF

### 2025 nvim-search-and-replace

Project-wide search and replace plugin for Neovim written in Lua: async streaming ripgrep search, live before/after preview, selective replacement and its own undo/redo stack.
github.com/mihaicristianfarcas/nvim-search-and-replace

Technologies Lua, Neovim, ripgrep


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## PERSONAL SKILLS

Mother tongue: Romanian

Other languages

| | UNDERSTANDING | | SPEAKING | | WRITING |
|---|---|---|---|---|---|
| | Listening | Reading | Spoken interaction | Spoken production | |
| English | C2 | C2 | C2 | C2 | C2 |
| C2 Certificate of Proficiency in English (Cambridge English Assessment) | | | | | |

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
Common European Framework of Reference for Languages

Digital skills

- Programming languages: Rust, Go, Python, Swift, C/C++, TypeScript/JavaScript, Lua, Shell
- Frontend: React, Next.js, Astro, Tailwind CSS, TLDraw, GPUI, Vercel, Cloudflare Workers
- Backend and data: FastAPI, PostgreSQL, Qdrant, FAISS, LangChain, PyTorch, LLM APIs
- AI and retrieval: RAG, hybrid dense/sparse/image retrieval, RRF fusion, reranking, citation grounding, Whisper, MCP
- Systems, networking and security: eBPF, WireGuard and the Noise protocol, ML-KEM-768 post-quantum key exchange, NAT traversal, BitTorrent DHT, Reed-Solomon FEC, Metal, Core Audio
- Tools and platforms: Git, Neovim, ripgrep, Linux, macOS, Windows


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